



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE
ICS 3103: AUTOMATA THEORY AND COMPUTABILITY
CAT 1

Date: 5th June 2025

Time: 1 Hours

Finite state machine- 3 story elevator problem

1) A new and improved elevator is installed in a 3-floor building. You are called upon to design the controller for the new elevator. The inputs (X,Y) for this controller are designated as follows:

XY	Operation
00	No request (stay on the same floor)
01	Move to first floor
10	Move to second floor
11	Move to third floor

The elevator requires *1 clock cycle* to travel between adjacent floors, however takes *2 clock cycles* to move from first floor to third floor and third floor to first floor. We could model this problem as follows:

State	Operation
A	Elevator is stopped on first floor
B	Elevator is stopped on second floor
C	Elevator is stopped on third floor
D	Elevator is moving between the first and third floor

While in state D, we assume that the elevator will only accept inputs (XY) 01 or 11 so that the next state after state D could only be A (first floor) or C (third floor). Include in your design 3 outputs, L1, L2, and L3. Each will connect to an LED which lights up when equal to 1 for floors 1, 2, and 3.

a) Is this state machine best implemented as a Mealy or Moore machine? Explain your reasoning.

b) Draw the state machine diagram that describes the operation of the elevator controller. Provide the state transition table corresponding to this state machine diagram.

[22 marks]